



**The University of Azad Jammu and Kashmir,
Muzaffarabad**

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How to connect two switches to form a LAN

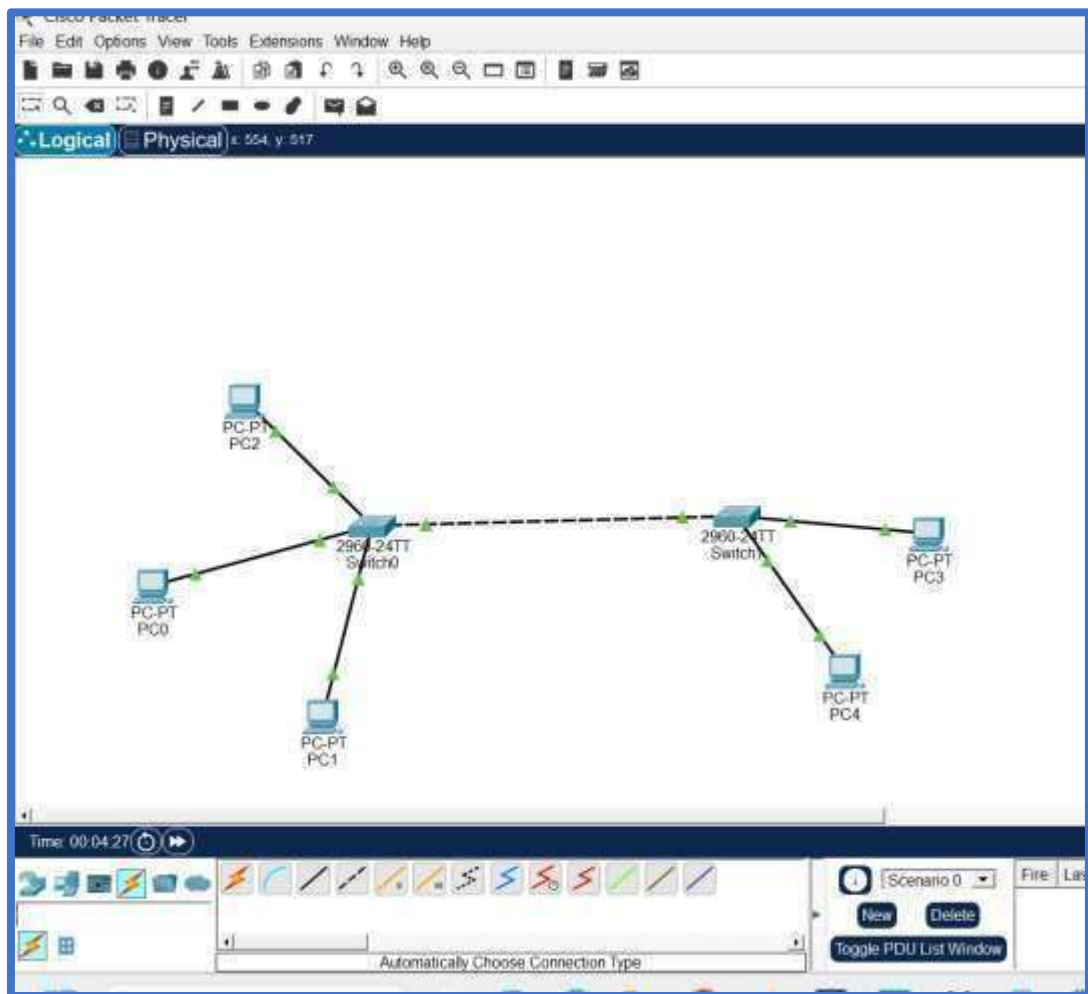
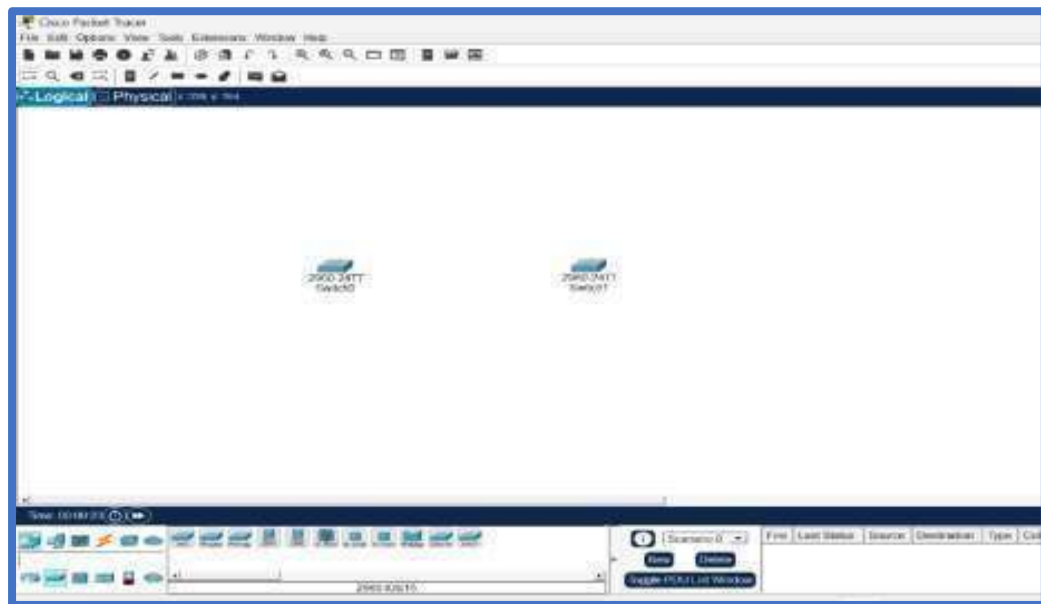
This is a **Star-Bus hybrid topology** consisting of two interconnected local network segments.

Key Components

- **Infrastructure:** Two **Cisco Catalyst 2960-24TT** switches linked together.
- **End Devices:** A total of **5 PCs** distributed across the switches (3 on Switch0, 2 on Switch1).
- **Connectivity:**
 - **Solid Black Lines:** Represent **Copper Straight-Through** cables used to connect the PCs to the switch ports.
 - **Dashed Black Line:** Represents a **Copper Cross-Over** cable used for the switch-to-switch connection (standard for connecting like-devices in older simulations).
- **Status Indicators:** All link lights are **Green (▲)**, indicating that the physical layer connections are active and the ports are currently in a "Forwarding" state.

Functional Summary

The configuration creates a single, extended **LAN (Local Area Network)**. Because the switches are bridged together, all 5 PCs are in the same broadcast domain, allowing them to communicate with one another once IP addresses are configured.



- **IP Configuration**

The screenshot shows a configuration window for a PC named 'PC0'. The 'Config' tab is active, and the 'Desktop' sub-tab is selected. The 'IP Configuration' section is expanded, showing settings for the 'FastEthernet0' interface. The 'IP Configuration' section has two radio buttons: 'DHCP' (unselected) and 'Static' (selected). Below these are fields for 'IPv4 Address' (192.168.1.1), 'Subnet Mask' (255.255.255.0), 'Default Gateway' (0.0.0.0), and 'DNS Server' (0.0.0.0). The 'IPv6 Configuration' section also has two radio buttons: 'Automatic' (unselected) and 'Static' (selected). Below these are fields for 'IPv6 Address' (empty), 'Link Local Address' (FE80:200:CFF:FE91:DC49), 'Default Gateway' (empty), and 'DNS Server' (empty). The '802.1X' section has a checkbox 'Use 802.1X Security' (unchecked). Below this are fields for 'Authentication' (MD5), 'Username' (empty), and 'Password' (empty).

PC0	
Physical Config Desktop Programming Attributes	
IP Configuration	
Interface FastEthernet0	
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address :	
Link Local Address	FE80:200:CFF:FE91:DC49
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.2

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::202:10FF:FE83:5A30

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC2

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.3

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80:230:F2FF:FE58:691E

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC3

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.4

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80:20D:8DFF:FE33:ECE1

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC4

Physical Config Desktop Programming Attributes

P Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.5

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::2D0:BCFF:FEA6:47B7

Default Gateway

DNS Server

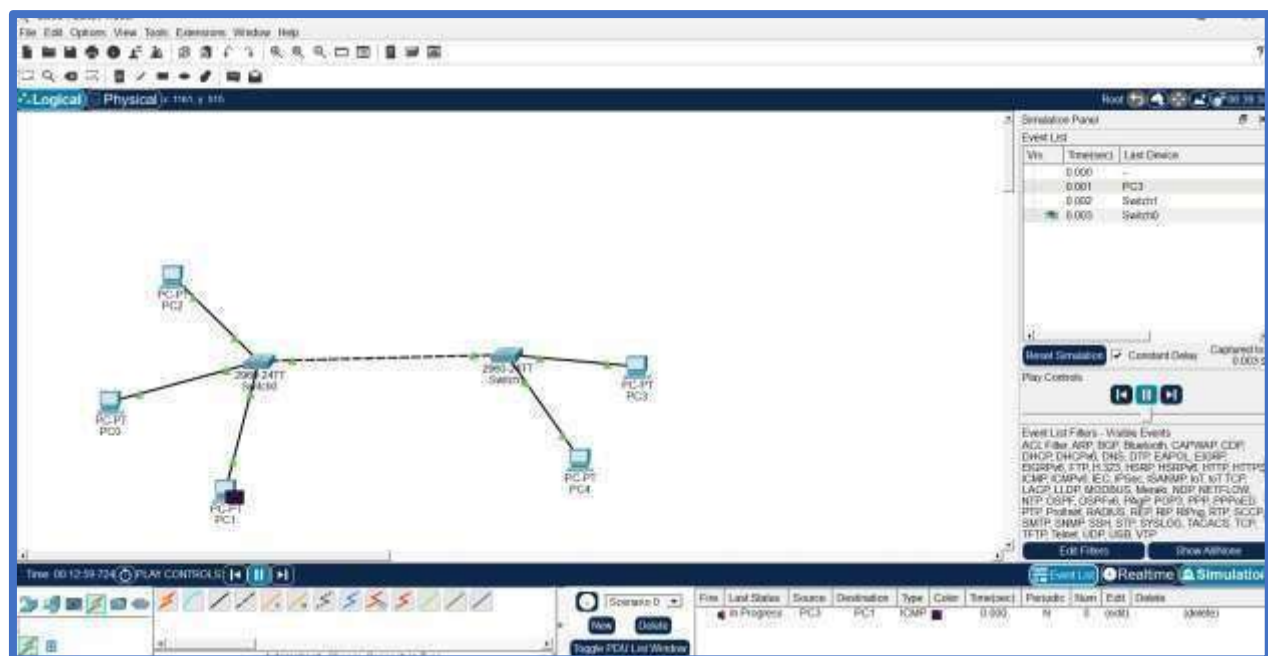
802.1X

☐ Use 802.1X Security

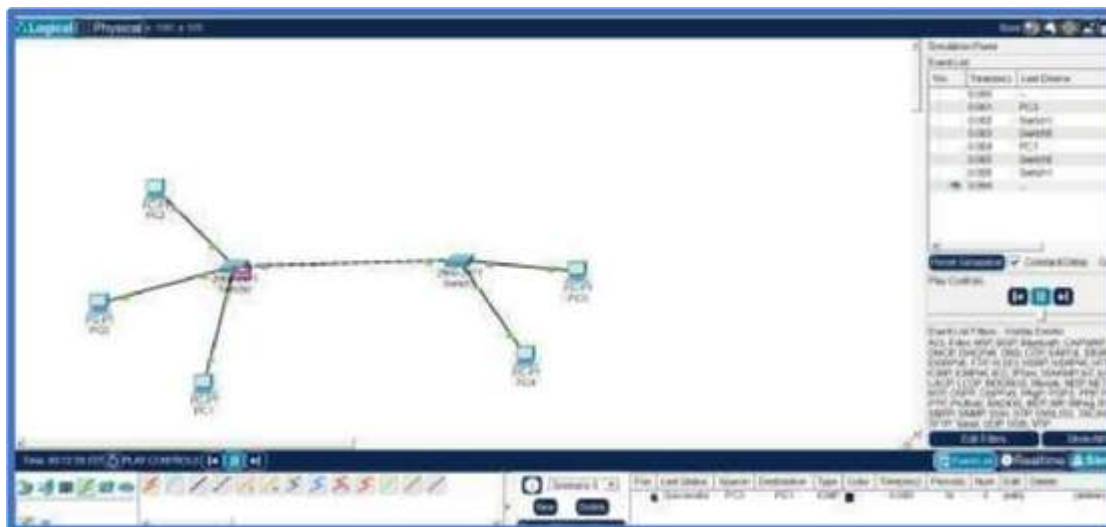
Authentication MD5

Username

Password



- **Network Topology:** The setup includes five PCs connected to two Cisco 2960-24TT switches. **Switch0** connects to PC0, PC1, and PC2, while **Switch1** connects to PC3 and PC4. The two switches are linked together via a dashed line, indicating a **Crossover Cable**.
- **Simulation Panel:** On the right, the "Event List" tracks network traffic in real-time. It shows a sequence of events starting from PC3, moving through Switch1, and arriving at Switch0.
- **PDU Information:** At the bottom, the "User Created PDU" window shows an active **ICMP (Ping)** request. The source is **PC3** and the destination is **PC1**, with the status currently "In Progress."
- **Visual Indicators:** The green triangles on the connection lines signify that the link lights are "up" and active, though a purple envelope icon on PC1 indicates it is the target of the current simulation packet.



Simulation Overview

- **Action:** An **ICMP (Ping)** packet is being sent across the network.
- **Visual Indicator:** A **purple envelope icon** is visible on **Switch0**, indicating the packet is currently in transit through that specific device.
- **Scenario Result:** The bottom right "PDU List" window shows a status of "**Successful**" for an ICMP traffic flow from **Source: PC3** to **Destination: PC1**.

Physical View :

